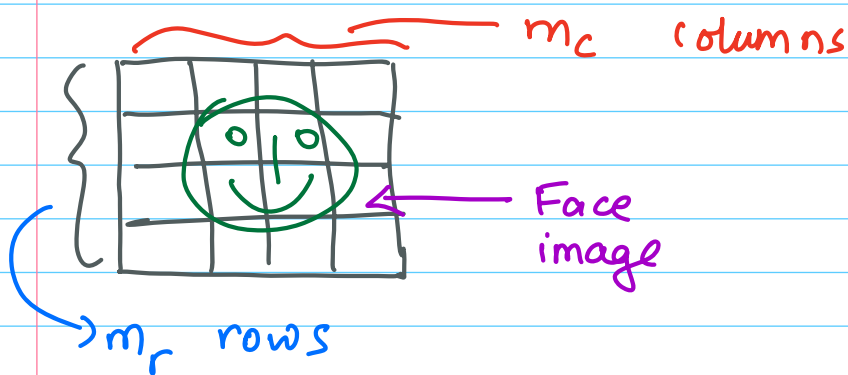


EIGEN FACES

Extension of
PCA for Face recognition

m_L

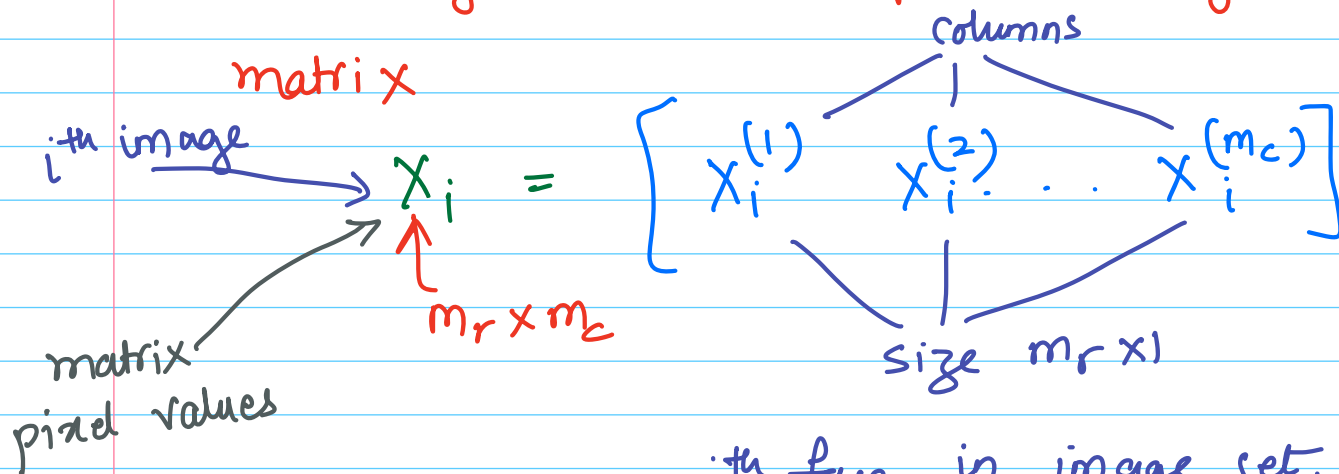


$m_c = \#$ of columns
of pixels

$m_r = \#$ of rows
of pixels.

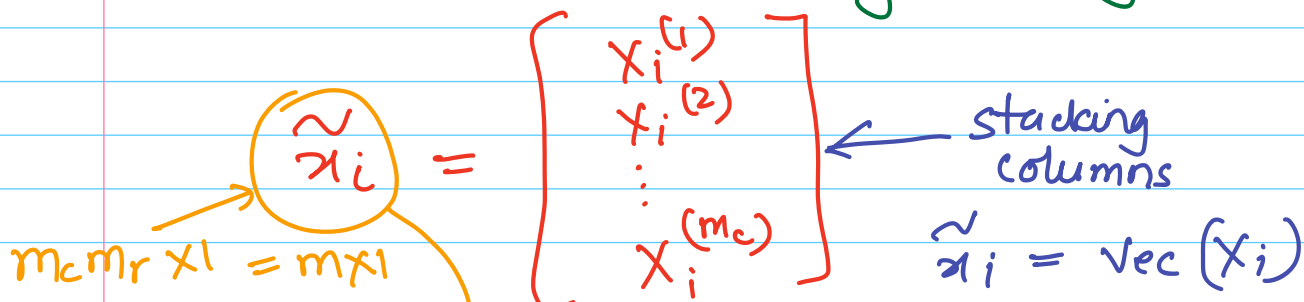
Total # pixels = $m_c \cdot m_r$

Each image can be represented by the



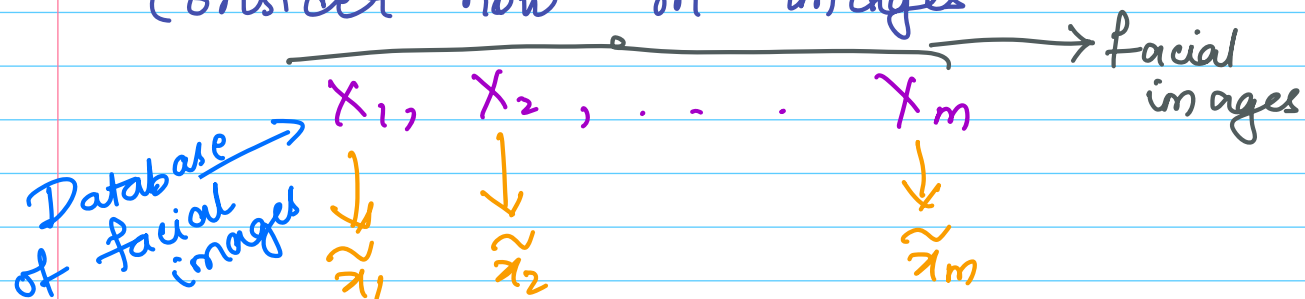
From X_i ← i -th face in image set.

obtain vector $\tilde{x}_i$ by stacking columns.



→ vector of pixel values.

Consider now m images



Given a facial image $\tilde{x}$, how to recognize?

Procedure for Eigenfaces :

Similar to PCA,

sample mean → $\bar{\mu} = \frac{1}{n} \sum_{i=1}^n \tilde{x}_i$

$\bar{x}_i = \tilde{x}_i - \bar{\mu}$ → subtract sample mean

Form covariance matrix

$R = \frac{1}{n-1} \sum_{i=1}^n \bar{x}_i \bar{x}_i^H$

$R = V \Lambda V^T$

Eigenvalue decomposition

Find eigenvectors

$\bar{v}_1, \bar{v}_2, \dots, \bar{v}_p$

Eigenvectors corresponding to

Largest principal components.

p largest eigen values of R

For each image $\bar{x}_i$, form the principal Component vector.

$$\bar{w}_i = \begin{bmatrix} \bar{v}_1^T \\ \bar{v}_2^T \\ \vdots \\ \bar{v}_p^T \end{bmatrix} \bar{x}_i$$

principal Components for i^{th} face $\rightarrow$ $\bar{w}_i = v^T \bar{x}_i$

weight vector for face i .

$v = \begin{bmatrix} \bar{v}_1 & \bar{v}_2 & \dots & \bar{v}_p \end{bmatrix}^{m \times p}$

$$\begin{array}{ccccccc} \bar{x}_1 & , & \bar{x}_2 & , & \dots & , & \bar{x}_n \\ \downarrow & & \downarrow & & & & \downarrow \\ \bar{w}_1 & & \bar{w}_2 & & \dots & & \bar{w}_n \end{array}$$

weight vectors for Facial images in the database.

FACIAL RECOGNITION :

Let $\tilde{x}_i$ be the given face image.

$$\bar{x} = \tilde{x} - \bar{\mu}$$

Now find the weight vectors corresponding to $\bar{x}$

$$\bar{w} = V^T \bar{x}$$

→ weight vector for unknown face.

→ Compare this with weight vectors

$$\bar{w}_1, \bar{w}_2, \dots, \bar{w}_n$$

of images in database.

$$d_i = \underbrace{\| \bar{w} - \bar{w}_i \|_2}_{\text{distance metric}}$$

→ Find face i with least distance

$$\Rightarrow \tilde{i} = \arg \min_i d_i$$

→ recognized face!

Eigen faces algorithm
for face recognition.

Compare $d_{\tilde{i}}$ with threshold ϵ

if $d_{\tilde{i}} > \epsilon$

This implies unknown face does not
correspond to any face image
in existing database.

else if $d_{\tilde{i}} \leq \epsilon$

Then it matches to face $\tilde{i}$.

Announce Face $\tilde{i}$.